

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 7 Winter 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTCOE705B - DEEP LEARNING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Deep Learning models require
 - A) Less data
 - B) High computational power
 - C) Fewer computations
 - D) No training
- 2 What happens if the learning rate is set too high in gradient descent?
 - A) The algorithm converges faster
 - B) The parameters may oscillate or diverge
 - C) The algorithm stops immediately
 - D) It has no effect
- 3 Which optimization algorithm adjusts the learning rate for individual parameters?
 - A) Stochastic Gradient descent
 - B) Momentum Based GD
 - C) ADAM
 - D) None of the above
- 4 What is stochastic gradient descent (SGD)?
 - A) Gradient descent on the entire dataset
 - B) Gradient descent using one data point at a time
 - C) Gradient descent with a fixed learning rate
 - D) Gradient descent without iterations
- 5 What is the purpose of dropout in deep neural networks?
 - A) Preventing overfitting by randomly deactivating neurons during training
 - B) Simplifying the architecture of the network
 - C) Reducing the number of neurons in the network
 - D) Removing outliers from the training data
- 6 Which of the following technique resembles the autoencoders used for dimensionality reduction?
 - A) PCA
 - B) Batch Normalization
 - C) Ensemble methods
 - D) Dropout
- 7 What is a filter/kernel in CNN?
 - A) Activation function
 - B) Loss function
 - C) Matrix used to detect patterns
 - D) Regularization Method

- 8 In denoising autoencoder noise is applied to
 A)Encoder's input B)Decoder's input
 C)Encoder's output D)Decoder's output
- 9 What is bias in machine learning?
 A)Error due to overly simplistic assumptions
 B)Error due to noise in the data
 C)Error from excessive complexity
 D)None of the above
- 10 Increasing the number of hidden units generally:
 A)Reduces model capacity
 B)Increases representational power
 C)Guarantees perfect accuracy
 D)increases bias
- 11 Which of the following neural network is used for handling time-series data?
 A) RNN (Recurrent Neural Network)
 B) CNN (Convolutional Neural Network)
 C) GAN (Generative Adversarial Network)
 D) Autoencoder
- 12 Which of the following is a major advantage of LSTM networks?
 A) Ability to capture long-term dependencies
 B) Reduced computational complexity
 C) Enhanced feature extraction from images
 D) Improved training speed

Q2. Solve the following.

- A) Explain the perceptron model .What are the limitations of perceptron model? 6
- B) Explain Gradient Descent algorithm for training a neural network 6

Q3. Solve the following.

- A) Explain Principal Component analysis for dimensionality reduction. 6
- B) Explain the backpropagation learning in detail. 6

Q4. Solve Any Two of the following.

- A) List and explain the various activation functions used in modeling of artificial neuron.A neuron receives inputs $x=[2,1]$ weights $w=[0.3,0.6]$ and bias $b=0.2$. Compute the neuron output with ReLU activation. 6
- B) What is regularization? Explain L1 and L2- regularization 6
- C) What are autoencoders ? Explain in detail. 6

Q5. Solve Any Two of the following.

- A) Explain working of all the layers in Convolutional Neural Network 6
- B) Differentiate between LeNet and AlexNet architecture 6
- C) Discuss some key applications of Convolutional Neural Network and Artificial Neural Network. 6

Q6. Solve Any Two of the following.

- A) Explain Recurrent neural network architecture in detail. 6
- B) How Deep learning differs from machine learning.Explain in detail. 6
- C) Write a note on i)Data Augmentation ii) Batch Normalization 6

*** End ***